Bibek Koirala

Computer Vision & Machine Learning Engineer
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Summary

Computer vision and machine learning engineer completing an M.S. in Computer Science (4.0 GPA, expected Dec 2026). Built end-to-end object detection and instance segmentation pipelines on real-world, custom-annotated image data, from dataset curation and annotation through training, evaluation, and inference (mIoU 0.91–0.98, mAP@0.5 ≈ 0.95). First-author paper at IEEE ICMLA 2025 (oral), plus EMNLP 2026 and *Journal of Food Protection* publications. Also builds privacy-first ML applications that run entirely on the user's device, deploying detection, segmentation, speech recognition, and text-to-speech models in the browser with ONNX Runtime Web, WebGPU, and WebAssembly. Brings 5+ years of software R&D delivering real-time, mission-critical numerical software.

Education

Southern Illinois University Carbondale – M.S., Computer Science, GPA 4.0/4.0 Aug 2024 – Dec 2026

- Coursework: Deep Learning, Machine Learning and Soft Computing, Artificial Intelligence, Data Mining, Natural Language Processing, Parallel and Distributed Computing, Advanced Database Systems

Technical Skills

Computer Vision: Object Detection, Instance and Semantic Segmentation, Image Classification, Mask R-CNN, YOLO, CNNs, ResNet, Vision Transformers, Transfer Learning and Fine-Tuning, OpenCV, MediaPipe, OCR (Tesseract),
Deep Learning & ML: PyTorch, TensorFlow, scikit-learn, Transformers, Vision-Language Models, Speech Recognition, Text-to-Speech, Statistics, Convex Optimization, NumPy, SciPy, Pandas, Matplotlib
On-Device & Web ML: ONNX / ONNX Runtime Web, WebGPU, WebAssembly, Transformers.js, Model Quantization, Web Workers, Progressive Web Apps, IndexedDB, WebRTC
Data & MLOps: Dataset Curation, Annotation with CVAT, Annotation Formats (COCO, YOLO, Pascal VOC), Training and Evaluation Pipelines, Experiment Tracking, Model Versioning, Docker, CI/CD (GitHub Actions), Git, Linux
Generative & Agentic AI: LangGraph, LangChain, Retrieval-Augmented Generation (RAG), FAISS and Vector Databases, HuggingFace, Vision Transformers, LLMs
Languages & Databases: Python, C++, C, Java, JavaScript, Node.js, HTML/CSS, SQL, PostgreSQL

Experience

Graduate Research Assistant, Computer Vision – Southern Illinois Univ. Carbondale Aug 2024 – May 2026
Department of Computer Science, Carbondale, IL | Funded by USDA-NIFA Grant No. 2024-70001-43667

- Built an end-to-end instance segmentation and classification pipeline in PyTorch for time-stamped microscopy images of bacterial cultures, covering data curation, CVAT annotation, training, evaluation, and inference; achieved mIoU of 0.91 to 0.98 across growth stages.
- Designed a background-aware approach that stays robust across varying imaging backgrounds, enabling detection of *E. coli* and *Salmonella* Typhimurium after 2 hours of incubation (previous best: 3 hours) at mAP@0.5 of approximately 0.95; first-author paper and oral presentation at IEEE ICMLA 2025.
- Contributed deep learning methods to a phase-contrast imaging study for rapid *Salmonella* detection, published in the *Journal of Food Protection* (2026).
- Co-authored an EMNLP 2026 multilingual benchmark for localizing generative image edits, working with vision-language and generative models.
- Released the framework as open source (github.com/vbek/bacteria-detection) and presented results to interdisciplinary computer science and food science audiences.

Officer – Software Development and R&D, Nepal Govt May 2018 – Dec 2023
Kathmandu, Nepal

- Designed, built, and deployed a fire-control application (UI, computation engine, validation layer); reduced targeting error from 50 m to 10 m at 1000 m range and cut computation latency from minutes to microseconds by optimizing

numerical computation pipelines.
- Led technical design and delivery of mission-critical software from requirements through deployment, acting as architect, developer, and point of contact for end users across multiple units.
- Delivered a full-stack application (dashboard, application logic, relational database) for leave and training records, replacing paper workflows across multiple units.
- Built decision-support software computing logistical requirements for Bailey bridge deployment under real-world constraints, with technical documentation and user guidance for each system.

Publications

- **Koirala, B.**, Alsobeh, A. M., Dhahir, N., and AbuGhazaleh, A., "Background-aware instance segmentation for early detection of *E. coli* and *Salmonella* in time-stamped microscopy images," *IEEE ICMLA 2025*, pp. 1354–1361. Oral presentation. ieeexplore.ieee.org/document/11471447
- "Edits in the Wild: A Multilingual Benchmark for Localizing On-Device Generative Image Edits on Consumer Platforms," **EMNLP 2026**, Main Conference. Co-author.
- Dhahir, N., Rababa, M., **Koirala, B.**, AbuGhazaleh, A., and AlSobeh, A., "Rapid detection of *Salmonella* Typhimurium microcolonies using phase-contrast imaging and deep learning," *Journal of Food Protection*, p. 100934, 2026.

Selected Projects

LocalLabel: In-Browser ML Annotation Tool with AI Assist – bibek-koirala.com.np/tools/locallabel

- Built a fully client-side image and video annotation tool (boxes, polygons, masks, keypoints, keyframe-interpolated video tracks, TIFF/GeoTIFF support) as a private, install-free alternative to CVAT, exporting COCO, YOLO, CVAT, and Pascal VOC.
- Implemented model-agnostic AI pre-labeling: runs any user-supplied ONNX model on WebGPU with WebAssembly fallback in a Web Worker, and decodes YOLO, YOLO-seg, semantic, instance-mask, and Mask2Former-style outputs into editable annotations.

On-Device AI Tools Suite – bibek-koirala.com.np/tools

- **Document Reader:** offline, multilingual text-to-speech for PDF, Word, EPUB, PowerPoint, Excel, and scanned files, serving 4 ONNX TTS model families (Kokoro-82M, Piper, KittenTTS, Supertonic) with custom grapheme-to-phoneme processing, up to 31 languages and 3× playback speed.
- **Media Studio:** on-device Whisper subtitle generation via Transformers.js (WebGPU with a 4-bit quantized decoder, or WebAssembly), plus batch image and video conversion, compression, and trimming.
- **AR Survival:** webcam-controlled game driven by real-time MediaPipe face tracking, with each vision model isolated in its own worker to keep tracking latency low.
- Also built a PDF editor (pdf.js, pdf-lib), a receipt-scanning finance planner (Tesseract OCR), and multiplayer board games over peer-to-peer WebRTC with three.js 3D rendering; all tools process data locally with no uploads.

Plant Disease Detection: CNN vs. ResNet vs. ViT – github.com/vbek/Plant-Disease-Detection

- Benchmarked a custom CNN (34.7M params), fine-tuned ResNet50 (9.9M trainable params), vanilla ResNet, and Vision Transformer on PlantVillage, reaching 97.9% to 99.6% accuracy and weighing accuracy against model size for deployment.

SEEKBOT: Agentic RAG Platform – github.com/vbek/SEEKBOT

- Built a full-stack RAG application as a 10-node LangGraph agent workflow with a local FAISS vector store, local sentence-transformer embeddings, and HuggingFace Inference API access to Llama 3.1 8B and Qwen 2.5 72B; supports semantic search over PDF, DOCX, CSV, Markdown, and web content.

News Credibility Detection – github.com/vbek/News-Credibility

- Built an NLP and ML framework classifying articles using linguistic, semantic, and source-based features; achieved 97% precision and 98% recall on benchmark datasets.